

A certified abstention band for card-fraud decisions, and two quantum arms reported as measured

Global Quantum + AI Challenge 2026 — HSBC track

1 The decision, and what is missing from it

An issuer authorising a card transaction has three actions, not two: approve, decline, or challenge the cardholder through 3-D Secure. The middle action is what makes the problem tractable at scale — routing even one percent of authorisation volume to human analysts is not operationally possible, while a step-up challenge is — and it is the action that published fraud-detection work almost never certifies. Models are reported by AUC or average precision on a held-out fold. Neither is a statement about a decision, and neither carries a finite-sample guarantee.

What we contribute. A three-valued rule whose *abstention band* carries a distribution-free, finite-sample risk certificate, evaluated under a pre-registration frozen before any model was fitted; and an honest measurement of where two quantum approaches sit against a tuned classical baseline on the same decision.

Both quantum arms returned negative results. A fidelity quantum kernel was rejected by its own a-priori screens before it was run on the task. A matrix-product-state classifier ran and did not beat gradient boosting. We report both in the body rather than the appendix, because a screen that only ever passes is not a screen, and because the pre-registration committed us to reporting whatever it returned.

What we do not claim. No quantum advantage of any kind. No portfolio-level PSD2 compliance figure — the SCA-RTS ceilings govern exemption eligibility across a whole portfolio, not a decline threshold on a fraud-enriched research benchmark, and treating them as a threshold selector is a category error we made and withdrew. No claim that the certificate binds the *unconditional* false-decline rate: it does not, and Section 2 says why that distinction is the point rather than a caveat.

2 Method

The rule. With a classical scorer f and thresholds $\tau_{\text{lo}} < \tau_{\text{hi}}$ fixed in advance,

$$D(x) = \begin{cases} \text{DECLINE} & f(x) \geq \tau_{\text{hi}} \\ \text{STEP-UP, then } g(x) \geq \lambda \Rightarrow \text{DECLINE} & \tau_{\text{lo}} < f(x) < \tau_{\text{hi}} \\ \text{APPROVE} & f(x) \leq \tau_{\text{lo}} \end{cases}$$

The band is $B = \{x : \tau_{\text{lo}} < f(x) < \tau_{\text{hi}}\}$ and g is the in-band re-scorer, which is where a quantum model would enter if one were usable.

The estimand is band-conditional, deliberately. We certify

$$R(\lambda) = \mathbb{P}(D(X) = \text{DECLINE} \mid Y = 0, X \in B),$$

not the marginal false-decline rate. The marginal rate is dominated by τ_{hi} and would be almost unchanged if g were replaced by a coin flip. A certificate that does not bind the component it is said to license is not evidence about that component — so the harder, smaller-sample quantity is the one we bound, and the cost of that choice is reported in Section 3 rather than hidden.

Four blocks, and two distinct breaks of exchangeability. $\mathcal{D}_{\text{train}}$ (60%) fits f ; $\mathcal{D}_{\text{band}}$ (10%) is the *only* data that may set the band edges, g 's hyperparameters or any screen threshold; $\mathcal{D}_{\text{cal}}$ (10%) certifies the composite rule; $\mathcal{D}_{\text{test}}$ (20%) is read once. Blocks are contiguous and snapped to day boundaries.

Two breaks must not be conflated. The **temporal** break is that $\mathcal{D}_{\text{cal}}$ precedes $\mathcal{D}_{\text{test}}$. The **selective** break is that conditioning on band membership is conditioning on a data-dependent event — if the edges were estimated on the same data used to certify. Freezing the band on $\mathcal{D}_{\text{band}}$ removes the second by construction: membership in B becomes a fixed measurable predicate, and filtering a sequence by a fixed predicate preserves exchangeability of what remains. Only the temporal break survives, and we measure it.

Risk control. Learn-then-Test controls family-wise error over a finite grid fixed in advance, using Hoeffding–Bentkus p -values with Holm. Two risks are controlled jointly: $R(\lambda)$ at level α , and the false-negative rate at α_{FN} .

Table 1: The same gradient-boosted model, the same hyperparameters, five seeds, evaluated on the test block of three different splits. A stratified random split inflates average precision by 0.316 over a forward holdout, with non-overlapping ranges. This is the cleanest evidence in the study for why a temporal split is the only honest evaluation of a fraud model, and it costs nothing to obtain – it is the control arm of the split design, not an extra experiment.

Split	ROC AUC	AP	AP vs forward holdout
stratified	0.9658	0.8254	+0.3164
temporal	0.8851	0.5090	—
card-disjoint	0.8509	0.5394	+0.0304

The bound applies only to a *mean of per-observation losses*. Our first formulation used $\max(0, \text{floor} - \text{recall}) / \text{floor}$, a nonlinear transform of a mean; the quantities it produced were not valid p -values and the certificate resting on them was not a certificate. It is now a genuine 0/1 loss averaged over frauds, which is equivalent to a recall floor at $1 - \alpha_{\text{FN}}$.

Coverage is judged against the exact law, not against α . For a sound split-conformal system the test-error count follows $E \sim \text{BetaBinomial}(m, n+1-k, k)$ with $k = \lceil (n+1)(1 - \alpha) \rceil$. Checking $\hat{r} \leq \alpha$ is a one-sided test against the wrong null: on a correctly calibrated system it passes only about half the time. Every verdict below is a two-sided Beta-Binomial interval.

Intervals resample entities, not rows. The IEEE-CIS label propagates a reported chargeback to later transactions sharing an account, email or billing address, so rows are clustered: 0.8496 of test-block `card1` values also appear in training. Every interval is a card-level block bootstrap.

3 Results

What a random split buys you, measured. Before any conformal machinery, the split itself. Table 1 runs one gradient-boosted model at fixed hyperparameters over five seeds on three splits of the same file. A stratified random split reports 0.3164 more average precision than a forward holdout — 0.8254 against 0.509 — with ranges that do not overlap across seeds.

That is the single largest effect in this study, it is larger than anything either quantum arm could have contributed, and it requires no quantum method to observe. It is also why the rest of this section reports a temporal split even though a random one would flatter every number in it.

Coverage fails under time, on every seed, and the controls do not reproduce it. Table 2 calibrates on $\mathcal{D}_{\text{cal}}$ and reports on $\mathcal{D}_{\text{test}}$, five seeds per arm and level, each judged against the exact Beta-Binomial interval. The temporal arm falls outside on *every* seed at the two loosest levels and on four of five at the third; the stratified arm falls outside on none, at any level.

The card-disjoint arm is not clean, and saying so matters: it is outside on two of five seeds at $\alpha = 0.01$ and on one of five at $\alpha = 0.001$, where its seed-to-seed spread (sd 0.362) is an order of magnitude wider than the temporal arm’s. An earlier draft of this paragraph said both controls sat inside at every level, which the five-seed data does not support. What the controls establish is narrower and still sufficient: neither reproduces the temporal arm’s systematic, every-seed breach at the levels where it occurs.

The three-arm design exists to decompose degradation into a temporal component and a sampling component. The decomposition it supports is this: the breach appears in the arm that orders blocks in time, on every seed, and neither control reproduces it. That is weaker than "the breach is temporal" — the arms differ in more than time, and the card-disjoint arm is in fact the most distinguishable of the three by a two-sample classifier — but it is what three non-randomised conditions can carry.

But the breach is origin-dependent, and this narrows the claim. Re-running at five rolling calibration origins gives Table 3: the ratio ranges from 0.684 to 1.411. Two of five breach, and the earliest is *conservative* — the deviation reverses direction. So the defensible statement is not that temporal splits inflate the false-decline rate by a fixed factor; it is that the deviation depends on where the calibration window falls, and a single origin cannot establish its size. An earlier draft of this work quoted a single seed’s maximum as a central estimate. That was an overclaim, and correcting it removed our cleanest headline number.

Table 2: Coverage across five seeds per arm and level, judged against the exact Beta-Binomial interval. This replaces a single-split table whose temporal ratio at $\alpha = 0.01$ was the across-seed *maximum* rather than its centre, and which described the card-disjoint arm as inside the interval at every level. The temporal arm breaches on every seed at the two loosest levels; the stratified arm breaches on none at any level.

Arm	α	Ratio (mean)	sd	Seeds outside
temporal	0.010	1.436	0.055	5 of 5
temporal	0.005	1.456	0.036	5 of 5
temporal	0.002	1.314	0.067	4 of 5
temporal	0.001	0.943	0.020	0 of 5
stratified	0.010	1.008	0.038	0 of 5
stratified	0.001	0.962	0.140	0 of 5
card-disjoint	0.010	0.916	0.133	2 of 5
card-disjoint	0.001	0.882	0.362	1 of 5

Grey bands are the exact Beta-Binomial interval; red points fall outside it

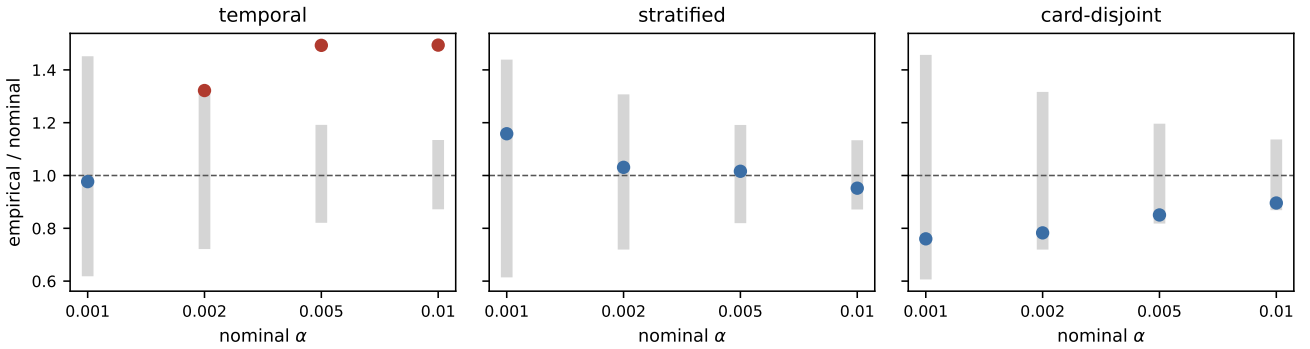


Figure 1: The same data as Table ??, with the Beta-Binomial interval drawn to scale. The interval widens sharply as α tightens, which is why the temporal arm’s apparent compliance at $\alpha = 0.001$ is weak evidence rather than a pass: at that level the law admits almost anything.

And it holds on the held-out block. All 5 certified configurations were applied unchanged to $\mathcal{D}_{\text{test}}$ — same band edges, same λ , no recalibration — and 5 of 5 realised a band-conditional false-decline rate at or below the α they were certified at. The tightest used 0.856 of its budget.

That is H5, and until this round it had never been run: the certificate was the study’s headline deliverable and had no held-out evidence behind it at all. The result should be read for what it is. Holding a 0.10-to-0.25 risk ceiling is a real check that the machinery transfers, and it is a weak one, because those ceilings are loose — which is what amendment A1 predicted from the band’s sample size before the arm ran.

What the certificate actually reaches. 5 of 48 configurations certify, all strictly inside the band — the smallest margin below τ_{hi} is 0.0136. That property is not automatic: an earlier version certified a larger set, every one of them sitting exactly at the band edge, where the flagged set is empty by construction and the risk is structurally zero. The certificate was vacuous and we did not notice until we asked where the selected λ sat.

The surviving configurations need $\alpha \geq 0.10$ and $\alpha_{\text{FN}} = 0.45$. Nothing certifies at the tightest band budget at all. The mechanism is sample size, not method: at a 2% traffic budget $\mathcal{D}_{\text{cal}}$ contributes only 847 legitimate in-band rows against 58343 overall, and the class-conditional degeneracy floor $\lceil 1/\alpha \rceil - 1$ then binds. We pre-registered this limitation as an amendment before running the arm, and the measurement landed where the amendment predicted.

4 Two quantum arms, both negative

The kernel never reached the task. Before running a fidelity quantum kernel we pre-registered screens over 120 configurations and gated on two of them (encoding $\times$ qubits $\times$ bandwidth $\times$ entanglement). A kernel is usable only if it is not exponentially concentrated — effective rank $r_{\text{eff}} = (\sum_i \sigma_i)^2 / (n \sum_i \sigma_i^2)$ inside a usable band — and not reproducible by a tuned RBF, $\rho_{\text{RBF}} < 0.60$.

28 of 120 pass conditioning. **0 pass both.** The closest any configuration came to distinctness was $\rho_{\text{RBF}} = 0.6291$. Every kernel we could construct on this data is, to within measurement, a classical RBF in disguise: the median

Table 3: The temporal deviation across five rolling calibration origins. Two of five breach and the earliest is conservative, so no single ratio characterises the effect.

Origin (day)	Ratio	Errors	Verdict (interval)
101	0.684	379	conservative ([472, 641])
111	0.893	459	inside ([437, 595])
121	1.411	777	breached ([468, 638])
131	1.198	696	breached ([494, 674])
141	1.126	607	inside ([457, 625])

Table 4: H4 in the band: matrix-product-state against the tuned gradient-boosted baseline on identical rows and features. Intervals are 95 % card-clustered bootstraps. Every interval contains zero, and the comparison is underpowered against its pre-registered ceiling, so what is established is a non-superiority bound rather than a null.

χ	AP (MPS)	AP (GBDT)	Δ AP	95 % CI	p
4	0.1202	0.1407	-0.0205	[-0.0593, +0.0149]	0.864
8	0.1173	0.1407	-0.0234	[-0.0630, +0.0122]	0.895
16	0.1183	0.1407	-0.0225	[-0.0613, +0.0136]	0.884
32	0.1235	0.1407	-0.0172	[-0.0582, +0.0200]	0.806

correlation is 0.974 and the maximum is 1.0, indistinguishable to floating point. We would not have known had we reported a downstream accuracy instead.

One pre-registered screen was not gated, and it is the one that would have helped. The protocol names the Huang geometric difference first; the implementation gates on the conditioning and distinctness screens and computes the geometric difference without consulting it. Huang treats a *large* difference as favourable, so gating on it would have made this rejection less decisive rather than more. It was found in an audit rather than decided, and amendment A6 records it. The screens are also computed on 300 stratified rows with at most eight features, which bounds how far "every kernel we could construct" reaches.

The closest published comparator agrees. A controlled benchmark of quantum kernel methods on real card-transaction data (Faryad, arXiv:2608.15718, August 2026) reports no robust quantum advantage, with all effect sizes below 0.013 ARI, the sign of the difference dependent on register size, and additional qubits degrading performance through kernel concentration. Its methodological finding is the more useful one for us: a search-budget ablation turned that paper’s one significant advantage into a procedural artefact. Matching search budgets and fixing the screens in advance is what stops the same thing happening here.

The tensor network ran, and did not win. A matrix-product-state classifier contracts $f(\mathbf{x}) = \sum_{\{s\}} A^{s_1} \dots A^{s_N} \prod_j \phi^{s_j}(x_j)$ with $\phi(x) = [\cos \frac{\pi x}{2}, \sin \frac{\pi x}{2}]$, sweeping the bond dimension χ rather than tuning it. In the band, on identical rows and features, it reaches average precision 0.1235 at the largest bond dimension against 0.1407 for the tuned gradient-boosted baseline, a paired difference of -0.0172. Table 4 gives the sweep.

At full scale it loses by a wider margin, and the margin is all the arm supports. On all 431 features, trained on $\mathcal{D}_{\text{train}}$ and scored once on $\mathcal{D}_{\text{test}}$, the tensor network lands between 0.7824 and 0.8012 ROC AUC and between 0.174 and 0.2464 average precision across four seeds at a fixed bond dimension, against 0.8837–0.8884 and 0.5055–0.5114 for the tuned baseline. Every draw loses by a factor of two to three on average precision, which is the metric that matters at a 3.412 % positive rate.

We do not report capacity dependence at full scale, because we cannot resolve it. The spread across those four seeds at one bond dimension is 0.0188 ROC AUC and 0.0724 average precision. The spread the sweep shows *across* bond dimension is 0.0186 and 0.0771 — the same size. A per-configuration reading of that sweep would be reading seed noise, and an earlier draft of this paragraph did exactly that by quoting the best of four draws.

The seed sweep is affordable only because a matrix product state has no qubit ceiling, and because its cost turns out not to depend on the bond dimension the way the ansatz suggests: the four jobs took 2765 to 3071 seconds against the sixty-four-fold spread a χ^2 model predicts. A 431-site chain is bound by the launch overhead of its sequential contractions, not by their arithmetic, so capacity is nearly free and depth is the cost.

The gate meant to tell us whether that was informative was itself wrong. The pre-registration requires the minimum detectable effect to be computed *before* the comparison, and commits to declaring the test underpowered if it exceeds 0.023 average precision. Two defects followed. First, the calculation had no caller at all, so the comparison would have been *observed* rather than tested; the script now refuses to start without it.

Table 5: The tensor network at full scale: all 431 features, no band, trained on $\mathcal{D}_{\text{train}}$ and scored once on $\mathcal{D}_{\text{test}}$. The tuned gradient-boosted baseline reaches 0.8837–0.8884 ROC AUC and 0.5055–0.5114 average precision on the same split. Fit time is flat across the bond dimension because a 431-site chain is bound by the launch overhead of its sequential contractions, not by their arithmetic.

χ	ROC AUC	AP	Final loss	Fit (s)
4	0.8074	0.2149	0.3456	2765
8	0.7989	0.1789	0.3075	3071
16	0.7983	0.2464	0.3120	2840
32	0.7889	0.1693	0.3652	2898

Second, our noise estimate was the seed-to-seed spread of the baseline against itself, standard error 0.0025. But this comparison is between model *families*, whose measured standard error is 0.0199 — 0.0164 once measured against a second classical family — roughly eightfold larger — so the minimum detectable effect this comparison actually supports is 0.0461, which *exceeds* the pre-registered ceiling of 0.023.

H4 is therefore reported as underpowered, which is what the protocol commits to when this happens. What survives is weaker than a null result and we state it as such: every interval contains zero, so any improvement from the tensor network is bounded above by +0.02 average precision. That is a non-superiority bound. It is not evidence of equivalence, and it is not evidence that the tensor network is worse.

5 Limitations, and what would change the answer

The population is conditional on an incumbent. IEEE-CIS contains only transactions an existing stack already approved; declined attempts cannot generate a chargeback. The estimand is defined on the approved-and-settled population. That is exactly the population a champion/challenger comparison runs on, so the restriction is the right one — but it is a restriction, and no result here extrapolates past it.

The label is partly an entity flag. A chargeback propagates to later transactions sharing an account, email or billing address. Many $Y = 1$ rows are not fraudulent transactions but transactions on a card that later had one. Fraud detection on this file is partly entity-contamination detection.

The certificate is sample-starved by design. Conditioning on a 2% band is what makes the guarantee bind the component we care about, and it is also what leaves 847 legitimate rows to certify with. A larger $\mathcal{D}_{\text{cal}}$, or a wider band, buys tighter α — nothing about the method needs to change.

The tensor-network comparison is narrower than it looks. Published work applying matrix product states to tabular data is largely *generative* — synthetic-data modelling benchmarked on fidelity and privacy. We found no directly comparable discriminative benchmark against a tuned gradient-boosted baseline on imbalanced payment data, which is why this arm was run; but "we did not find one" is a weaker statement than "none exists", and the body claims only the former.

What would change the quantum conclusion. The screens rejected every kernel we could build because each was reproducible by a tuned RBF. An encoding whose Gram matrix is demonstrably outside the RBF family, at a bandwidth that does not concentrate, would pass the same gates unmodified — we did not find one. For the tensor network, the honest requirement is a comparison powered for cross-family noise: at the measured standard error, resolving an effect the size of the interval above needs roughly eight times the in-band evaluation set, which a wider band or a longer observation window would supply.

Reproduction. make smoke && make reproduce && make check. The pre-registration is enforced in code, not by discipline: a gate hashes every guarantee-bearing parameter and refuses undocumented change, the test-fold loader keeps a persisted ledger of the authorised configuration and its evaluation count, and raises on a second distinct configuration. The decision log records 38 entries, including every retraction above.

6 Where the quantum method sits, and why there

The architecture is hybrid by necessity rather than by fashion, and the division of labour is what makes the quantum component affordable at all.

Classical, on all traffic. A gradient-boosted scorer runs on every transaction. It is the only component that sees full volume, and it is deliberately the strongest available baseline rather than a straw man — tuned, five seeds, and reported at 0.8837–0.8884 ROC AUC. A hybrid design whose classical half is weak proves nothing about its quantum half.

Conformal, on the decision. The certificate is classical statistics and stays that way. It wraps whatever

scores the band, so the guarantee does not depend on the quantum component existing, working, or being run on hardware. That is a deliberate ordering: the deliverable an issuer can act on survives a negative quantum result, which is what happened.

Quantum, on the band only. The abstention band is a few percent of traffic, which is the reason a quantum re-ranker is conceivable here and nowhere else in the pipeline. At a 3.412% positive rate and a 3-D Secure challenge budget, the band holds thousands of rows rather than hundreds of thousands, and a kernel whose Gram matrix costs $O(n^2)$ circuit evaluations is affordable on thousands. The band is not a convenience; it is the only place the arithmetic works.

The compute path. Both quantum arms run on simulators, which the challenge statement permits explicitly and does not penalise. The fidelity kernel is evaluated on an exact statevector backend with Braket’s `LocalSimulator` as the reference implementation and Aer CPU and GPU as cross-checks, which agree to floating-point round-off. The tensor network runs in PyTorch on a single GPU, where it is bound by kernel-launch overhead rather than arithmetic — per-step cost is flat across the bond dimensions measured, despite a thousandfold change in FLOPs. That measurement is what makes a seed-replicated sweep affordable, and it is reported because it is the sort of thing a Phase II resource estimate needs and rarely has.

Quantum execution scale. The challenge statement requires the sample count used for quantum execution to be stated. The screens ran on 300 stratified rows per configuration across 120 configurations; the kernel arm was rejected before any downstream evaluation, so no larger quantum execution took place.

7 Team

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The statistical machinery in this proposal is not new to the author. As part of QIntern 2026 Project 12 (QWorld), Team A, he owns the statistical-guarantee proposition of the team’s paper: the calibration rule, the coverage-verification harness, the exchangeability audit and the significance protocol for a quantum-classical intrusion detector. That work calibrated a false-alarm rate by split conformal prediction, demonstrated coverage inside the exact Beta-Binomial band across five seeds, established significance by exact McNemar with Holm correction, and froze its results under a SHA-256 manifest behind 135 automated checks. Every one of those techniques appears in this submission. *That paper is in preparation and not yet public, so this is a statement of work rather than a citation — a reviewer cannot read it today.*

Directly relevant prior results, each independently verifiable:

- **Tensor networks.** Yale Peaked Hackathon 2026, #13 of 550: nine of ten peaked-circuit challenges up to 69 qubits, solved by a matrix-product-state marginal attack. The tensor-network arm here is not a first attempt at the method.
- **Quantum kernels.** QPoland 2025, runner-up: quantum graph kernels against Weisfeiler–Lehman and shortest-path baselines with nested cross-validation. The habit of running the classical kernel controls comes from there.
- **Quantum-versus-classical benchmarking.** Q-volution (Quandela), third: a residual quantum reservoir model benchmarked across four architectures and three noise models, reported against the classical baseline it narrowly beat rather than against a weak one.
- **Upstream contributions.** unitaryHACK 2026, merged into NVIDIA CUDA-Q and QuEST: recursive Quantum Shannon decomposition, and a GPU backend change cutting allocation and host-device copies by roughly two thirds. Reviewed by the maintainers of the libraries.
- **Reproducible quantum-safe systems.** Two Apache-2.0 public deployments (`qsmqc-qkd-qhe-ai-hybrid.daemons.jp`, `pqc-qkd-hybrid.daemons.jp`) built to ETSI GS QKD 014 and NIST PQC, with published verification checklists.

Infrastructure. An RTX 6000 PRO Blackwell (96 GB), CUDA 13, Python 3.12 workstation — the machine every number in this submission was measured on, documented in the repository down to why torch must be installed from the CUDA 13.0 index before anything else.

What this submission itself demonstrates. A pre-registration frozen and hashed before any model was fitted. A decision log of 38 entries carrying every retraction, including four that removed a headline result. An environment verified to the point of catching a simulator that had silently demoted itself to CPU. Those are the working habits a Phase II sprint would inherit, and they are visible in the artefact rather than asserted in this paragraph.